

Tsung-Yi Ma

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Senior AI Engineer with a strong mathematical background (NTU Math; Phi Tau Phi Honor Society). Focused on Generative AI, real-time speech systems, and mathematical optimization. ASUS Global Role Model Award recipient.

EXPERIENCE

ASUSTEK COMPUTER INC.

Senior AI Engineer

AI Engineer

New Taipei City, Taiwan

Aug. 2025 – Present

Nov. 2023 – Aug. 2025

- **2025 Global Role Model Award:** Selected for the **Outstanding Contribution Award** (HQ selection, Top <1% company-wide) for exceptional performance and high-impact project delivery.
- **AI Interview Agent (GAI Recruiter):** Architected a real-time streaming speech recognition pipeline (currently in beta rollout) using **Python (asyncio)**, **Ray**, and **WebSockets** to serve **multiple concurrent interview sessions**. Implemented a **sliding-window approach** to sustain **sub-500 ms streaming latency**. Tuned VAD parameters to significantly reduce hallucinations.
- **Production Scheduling Optimization:** Designed large-scale mathematical models (MIP/LP, ~1M variables/constraints) using **CPLEX** and **SCIP** solvers for factory-wide production scheduling. Restructured constraints to cut solver runtime from **24+ hours to <1 hour (24× speedup)**. In simulation, the optimized schedule delivered **10% higher throughput** than a traditional FIFO baseline.
- **License Plate Voice Recognition:** Fine-tuned **OpenAI Whisper** for specialized alphanumeric recognition, boosting accuracy to **>80%**. Leveraged domain knowledge of real-world license plate formatting conventions to guide an **LLM-based post-processing** step that corrects ambiguous transcriptions. Implemented on-the-fly data augmentation (A-law encoding, noise injection) to ensure model robustness.

EDUCATION

National Taiwan University

Master of Mathematics

Taipei, Taiwan

Sep. 2021 – Aug. 2023

- **The Phi Tau Phi Scholastic Honor Society:** Inducted as an Honorary Member, an academic distinction typically awarded to top Master's graduates.
- **CS Foundations:** Data Structures & Algorithms (A+), Machine Learning (A).
- **Thesis:** "Introduction to the Ising Model and the Ising Model with Disorders". Applied rigorous analysis to statistical physics models.

National Taiwan University

Bachelor of Mathematics

Taipei, Taiwan

Sep. 2017 – Jun. 2021

- **Research:** Selected for **NCTS Undergraduate Summer Research Program** (Noncommutative Probability).

Taipei Municipal Chien Kuo High School

Science Class

Taipei, Taiwan

Sep. 2014 – Jun. 2017

- One of Taiwan's most selective pre-college science programs (~8% admission rate; 30 students/year nationally).

PROJECTS

Distributed Whisper Streaming | Python, Ray, faster-whisper

[GitHub](#)

- Added a **RayQueue**-based routing layer to share GPU-bound Whisper inference workers across concurrent streaming sessions.

Lox Tree-Walking Interpreter | Rust, recursive-descent parsing, AST evaluation

[GitHub](#)

- Implemented a tree-walking interpreter for the Lox language (*Crafting Interpreters*) – scanner, recursive-descent parser, and evaluator supporting closures, control flow, and first-class functions.

TECHNICAL SKILLS

Languages: Python (Primary); project experience with Rust, Go, and TypeScript; basic SQL

AI & Speech: PyTorch, OpenAI Whisper, faster-whisper, VAD, streaming inference

Optimization: Mathematical Programming, MIP/LP, CPLEX, SCIP

Backend & Systems: FastAPI, async Python, WebSockets, Ray, Linux

Developer Tools: Git, Ruff, Pyright